Lesson notes · for you

Distributions and histograms

Data science · Level 2 · Lesson 1

Time
30 min

Materials
A notebook and a pencil; A phone or tablet with Darsi, for the lab

Learning goals

- Build a histogram with equal bins.
- Read symmetric and skewed shapes.
- Choose a middle that fits the shape.

Activity: A sticky-note histogram

1. Warm-up (5 min): Guess whether your class's heights make a hill or a tail.
2. Measure (10 min): Measure your hand span and stick a note in the right bin on the board.
3. Read (10 min): With a partner, describe the shape and try it again with wider bins.
4. Wrap-up (5 min): Say your finding in one sentence with a number in it, then fill in the exit ticket.

Discussion

- Why must bins be equal?
- What did wider bins hide?
- Which middle fits a skewed shape?

Check yourself

- Sketch a skewed-right histogram and mark roughly where the mean and the median are.

Try it this week
Measure the hand spans of your family or friends, put them in equal bins and draw a histogram. Is it a hill, or does it have a tail?

Worksheet

Distributions and histograms

Data science · Level 2 · Lesson 1

Name _____ Class _____ Date _____

1 **In a histogram of test marks, what does the height of each bar show?**

◯ How many pupils have marks in that bin ◯ The average mark ◯ The best mark

2 **In a histogram, the height of a bar is a count.**

◯ True ◯ False

3 **Count the heights into five 5 cm bins starting at 140, and print the counts.**

```
h = [142, 147, 151, 153, 156, 158, 161, 149, 152, 155]
c = [0, 0, 0, 0, 0]
for x in h:
    c[0] = c[0] + 1
print(c)
```

Write your program:

4 **Match each dataset to its likely shape.**

Heights of 13-year-olds	• • • •	Skewed right
Minutes on a phone each day	• • • •	Two hills
Heights of pupils and their parents together	• • • •	Symmetric

5 **Phone minutes are skewed right. Where is the mean?**

◯ Above the median ◯ Below the median ◯ Exactly on it

✎ **Which numbers in your life do you think are skewed, with a long tail?**

Check your answers

Distributions and histograms

Data science · Level 2 · Lesson 1

1 **How many pupils have marks in that bin**

Each bar counts the pupils whose marks fall in its bin.

2 **True**

Each bar counts how many values fall in its bin.

3

```
h = [142, 147, 151, 153, 156, 158, 161, 149, 152, 155]
c = [0, 0, 0, 0, 0]
for x in h:
    b = (x - 140) // 5
    c[b] = c[b] + 1
print(c)
```

Each height becomes a bin number from 0 to 4.

4 **Heights of 13-year-olds — Symmetric**
Minutes on a phone each day — Skewed right
Heights of pupils and their parents together — Two hills

Two very different groups mixed together make two hills.

5 **Above the median**

The long tail of big values pulls the mean up.